CLASS 10 — SCIENCE · FORMULA SHEET

Chapter 1: **Chemical Reactions and Equations** | Quick revision, one page

Chemical equations

- Reactants $\rightarrow$ products; a balanced equation has the same number of atoms of each element on both sides (law of conservation of mass).
- Physical states: (s), (l), (g), (aq). Condition (heat, catalyst, light) is written over the arrow. $\uparrow$ = gas evolved, $\downarrow$ = precipitate.
- Balancing: do not change formulas; change only coefficients.

Types of reactions

- Combination: $\mathrm{A + B \longrightarrow AB}$, e.g. $\mathrm{CaO + H_2O \longrightarrow Ca(OH)_2}$ (exothermic, slaking of lime).
- Decomposition: $\mathrm{AB \longrightarrow A + B}$; thermal: $\mathrm{CaCO_3 \longrightarrow CaO + CO_2}$; electrolytic: $\mathrm{2\,H_2O \longrightarrow 2\,H_2 + O_2}$; photolytic: $\mathrm{2\,AgCl \longrightarrow 2\,Ag + Cl_2}$.
- Displacement: $\mathrm{A + BC \longrightarrow AC + B}$, e.g. $\mathrm{Fe + CuSO_4 \longrightarrow FeSO_4 + Cu}$ (A more reactive than B).
- Double displacement: $\mathrm{AB + CD \longrightarrow AD + CB}$, e.g. $\mathrm{Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2\,NaCl}$ (precipitation reaction).

Oxidation and reduction

- Oxidation = gain of oxygen / loss of hydrogen. Reduction = loss of oxygen / gain of hydrogen.
- Redox: both occur together, e.g. $\mathrm{CuO + H_2 \longrightarrow Cu + H_2O}$ (CuO reduced, $\mathrm{H_2}$ oxidised).
- The substance oxidised is the reducing agent; the substance reduced is the oxidising agent.

Energy changes

- Exothermic: heat released (burning, respiration, slaking of lime, decomposition of vegetable matter).
- Endothermic: heat absorbed (decomposition of $\mathrm{CaCO_3}$, photosynthesis, electrolysis).

Key reactions to remember

- $\mathrm{2\,Pb(NO_3)_2 \longrightarrow 2\,PbO + 4\,NO_2 + O_2}$ (brown fumes of $\mathrm{NO_2}$).
- $\mathrm{2\,FeSO_4 \longrightarrow Fe_2O_3 + SO_2 + SO_3}$ (green crystals to brown residue).
- $\mathrm{Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O}$ (lime water turns milky).

Corrosion and rancidity

- Corrosion: slow attack on metals by air, moisture or acids (rusting of iron needs both air and water; green coating on copper, black coating on silver).
- Prevention: painting, oiling, galvanising, alloying. Rancidity: oxidation of fats and oils giving bad smell and taste; prevented by antioxidants, airtight containers, flushing with nitrogen, refrigeration.